

K : number of clients

T : current round/time step

k : current client to be starting compression and transmission

$g_{i,l}^t$: the l -th element of the gradient vector at worker i and time step t

$enc_i^t(??)$: the encoder at worker i , and used at time step t , which takes some input to encode each element $g_{i,l}^t$

$dec^t(??)$: the decoder at the server, used at time step t

$Aggr_l^t$: the gradient element aggregated by the server at time step t

$\phi(??)$: The training process happening on server

$g_{k,l}^T$: the current element that we want to encode, transmit and then decode

enc_i^T, dec^T : the encoder and decoder used for this compression

enc and dec uses:

SI : symbol used to signify a side info, without specifying exactly what.

$u_{k,l}^t = enc_k(g_{k,l}^t)$: is the the symbol for the encoded element $g_{k,l}^t$ at worker k , (the quantized value)

$\bar{g}_{i,l}^t = dec(u_{k,l}^t, SI)$: the l -th element of the reconstructed gradient vector at worker k and time step t , decoded by the server.

Variants of SI – the distinguishing factor:

- 1- $[\bar{g}_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$: i.e. All reconstructions so far
- 2- $[g_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$: i.e. all raw gradients (used only as oracle/perfect SI)
- 3- $[\bar{g}_{k,l}^y], y \in [0, T]$: i.e. reconstructions of the worker k

Training:

$[enc_k^t, dec^t] = \phi_k^t(X, SI)$: the training process, taking X as reconstruction target, and SI as side info used (from above list).

Variants of X – the second distinguishing factor:

- 1- $\bar{g}_{k,l}^{T-1}$: The last reconstructed l-th element for client k, from last round/time T-1
- 2- $g_{k,l}^T$: The current raw gradient l-th element, at the current time T, from worker K
- 3- $\bar{g}_{k,l}^T$: The reconstructed l-th element from client k, at current time T

Prior rate complication:

Rates reported are from the posterior estimation. The prior rate, so an implicit assumes a perfect SW encoding step on the full quantized vector $[u_{k,l}^T] l \in [0, N]$. Unless explicitly stated, the SI for SW is the same as the one used in training.

setup I want to talk about

1- Oracle case of Wyner-Ziv quantizer – the impossible and perfect case:

$$[enc_k^T, dec^T] = \phi(g_{k,l}^T, SI) \text{ and } SI = [g_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$$

We train on the lossless, raw gradient $g_{k,l}^T$ which is also going to be transmuted by the same enc and dec. the SI, is all past gradients, in their raw lossless form. This is of course an impossible case for any practical solution and is only an oracle.

2- Partial Oracle case of Wyner-ziv:

$$[enc_k^T, dec^T] = \phi(g_{k,l}^T, SI) \text{ and } SI = [\bar{g}_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$$

Slightly more practical case, much like above, but instead of raw lossless grads for SI, we use their reconstructions

2- Practical case of Wyner-ziv – the proposed algorithm that I worked on:

$$[enc_k^T, dec^T] = \phi(\bar{g}_{k,l}^T, SI) \text{ and } SI = [\bar{g}_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$$

its a bit of a complicated algorithm, so I won't go into details of how we do it. It replaces the raw gradients from case 1 with their reconstructions.

3- Practical case of Temporal – forget wyner-ziv, only see same clients past:

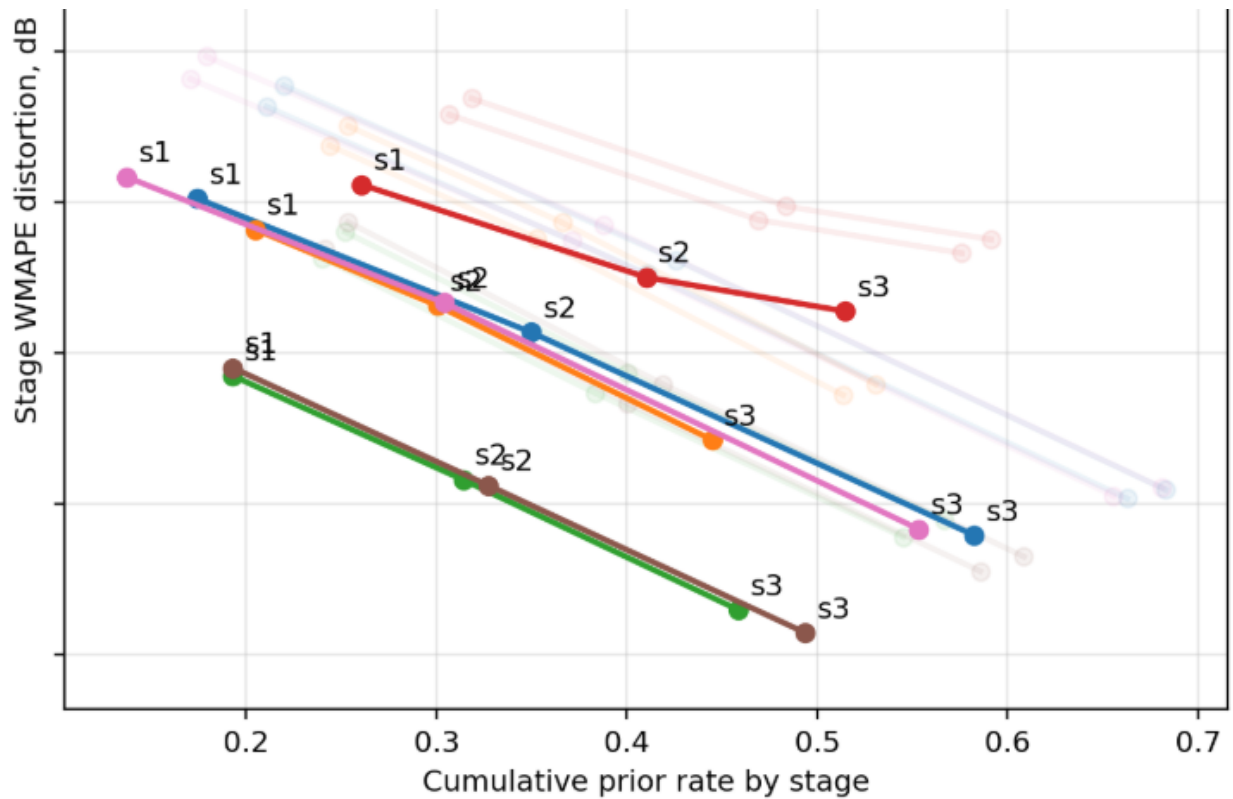
$$[enc_k^T, dec^T] = \phi(g_{k,l}^T, SI) \text{ and } SI = [\bar{g}_{k,l}^y], y \in [0, T]$$

We train on the lossless, raw gradient $g_{k,l}^T$ which is also going to be transmuted by the same enc and dec. the SI, is only the past recons of the same client.

4- Oracle case of Temporal – perfect setup for temporal case:

$$[enc_k^T, dec^T] = \phi(g_{k,l}^T, SI) \text{ and } SI = [g_{k,l}^y], y \in [0, T]$$

same as 2, but using raw gradients for SI, instead of reconstructions.



1- conclusion: adding other clients gradients to SI, has no effect. Knowing the differences is enough. Evidence:

Brown: Oracle case of Temporal training

Green: Oracle case of Wyner-Ziv case – the impossible and perfect case

2-The practical, well studied temporal solution, is only very slightly off from the partial oracle of WZ. Evidence:

Blue: Practical case of Temporal

Orange: Partial Oracle case of Wyner-ziv

3- The tiny gap from above (between blue and orange) is from SW and not the wz quantization. Evidence:

Pink: Practical case of Temporal, but the posterior was retrained with

$$SI_{PosteriorOnly} = [\bar{g}_{x,l}^y], x \in [0, K], y \in [0, T], (x, y) \notin [[k, K] \cdot [T]]$$

Final Conclusion:

Even a perfect version of WZ quantizer is a bad idea.

Too bad I actually spent weeks on making a near perfect implementation of it :(